

UWT-Net: Mining low-frequency feature information for medical image segmentation

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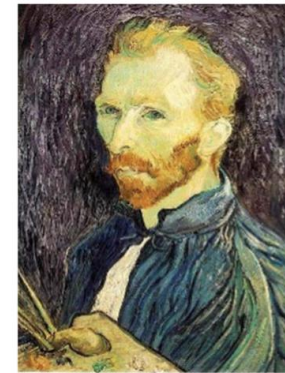
Abstract

Medical image segmentation is the core technology of precision medicine, which can improve diagnostic accuracy, optimize treatment plans, and enhance research efficiency. U-Net is a classical and fundamental model in this field. Because of its excellent architecture, Transformer and MLP have been fused on top of it in subsequent work, all with good results. Each of these methods has advantages, but none further explores the image's low-frequency feature information. The low-frequency feature information reflects the overall structure and contour of the image and provides key background and boundary information for image segmentation. To address this problem, we explore the potential of Wavelet Convolutions for medical segmentation tasks by proposing a novel feature extraction block: the Image Multi-frequency Feature Information Extraction (IMFIE) block. The IMFIE block can effectively extract both high-frequency and low-frequency feature information from images by combining Wavelet Convolutions. This approach takes full advantage of their excellent ability to mine and utilize low-frequency information in images while expanding the receptive field at a low cost. We propose a novel model, UWT-Net, which leverages the IMFIE block and reconstructs the classical U-Net. Experiments on three public pathology image datasets show that the proposed method outperforms the state-of-the-art baseline U-KAN. Code is available at <https://github.com/zpc2002zpc/UWT-Net.git>.

- Introduce
- Method
- Experiment
- Conclusion
- Review

- Medical image segmentation model
 - CNNs: U-Net, U-Net++ ... etc
 - CNNs are struggling to model global dependencies, resulting in suboptimal performance.
 - Attention: Att-Unet, TransUNet, Swin-Unet ... etc
 - The attention mechanism can effectively compensate for the disadvantage of CNNs.
 - Wavelet Transform: Spectral U-Net, Wet-unet
 - However, existing methods have demonstrated insufficient attention to low-frequency information in the images and do not further mine and utilize the low-frequency information.

- High-frequency information:
 - Edges, noises
 - Easily lost when down sampling
- Low-frequency information:
 - Global shapes, Spatial informations
 - Robust to noise



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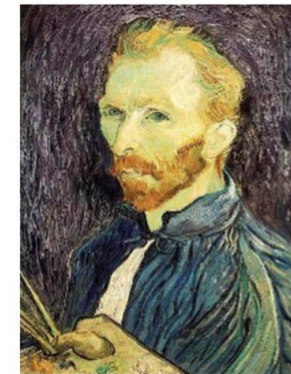


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1/4



1/2



1/4 (2x zoom)



1/8 (4x zoom)

<https://velog.io/@richpin/Computer-Vision-03-Image-Pyramid>

- Contribution
 - UWT-Net
 - Image Multi-Frequency Information Extraction Block
 - Conv+**WTConv**(*Shahaf E. Finder et al. / ECCV, 2024*)
 - SOTA + ablations

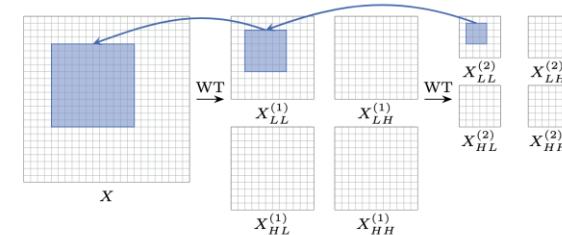


Fig. 2: Performing convolution in the wavelet domain results in a larger receptive field. In this example, a 3×3 convolution is performed on the low-frequency band of the second-level wavelet domain $X_{LL}^{(2)}$, resulting in a 9-parameter convolution that responds to lower frequencies of a 12×12 receptive field in the input X .

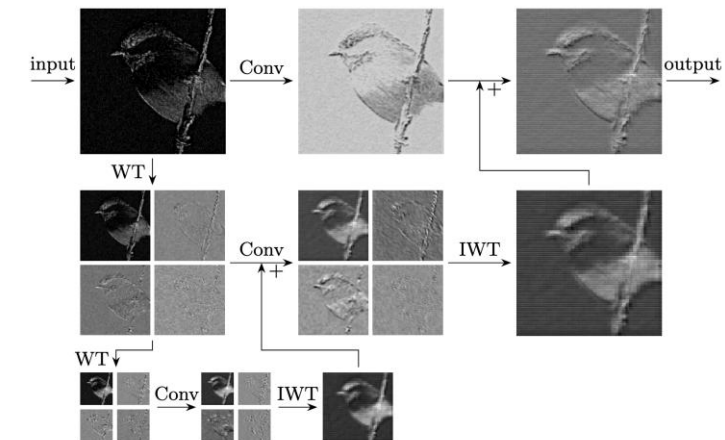


Fig. 3: An example of the WTConv operation on a single channel taken from the third inverted residual block of MobileNetV2 (see § 4.4) using a 2-level wavelet decomposition and 3×3 kernel sizes for the convolutions.

- Wavelet Convolutions
 - 2D Haar Convolutions

$$F_{LL} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad F_{LH} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix},$$

$$F_{HL} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}, \quad F_{HH} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$[N_{LL}, N_{LH}, N_{HL}, N_{HH}] = \text{Conv}([F_{LL}F_{LH}, F_{HL}, F_{HH}], N)$$

$$N = \text{Conv-transposed}([F_{LL}F_{LH}, F_{HL}, F_{HH}], [N_{LL}, N_{LH}, N_{HL}, N_{HH}])$$

$$N_{LL}^{(i)}, N_{LH}^{(i)}, N_{HL}^{(i)}, N_{HH}^{(i)} = \text{WT}(N_{LL}^{(i-1)})$$

$$N_H^{(i)}, N_{LL}^{(i)} = \text{WT}(N_{LL}^{(i-1)})$$

$$M_H^{(i)}, M_{LL}^{(i)} = \text{Conv}(W^{(i)}, (N_{LL}^{(i)}, N_H^{(i)}))$$

$$L^{(i)} = \text{IWT}(M_{LL}^{(i)} + L^{(i+1)}, M_H^{(i)})$$

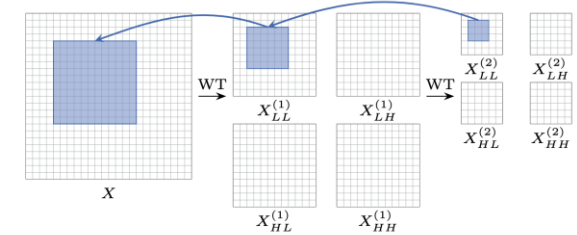


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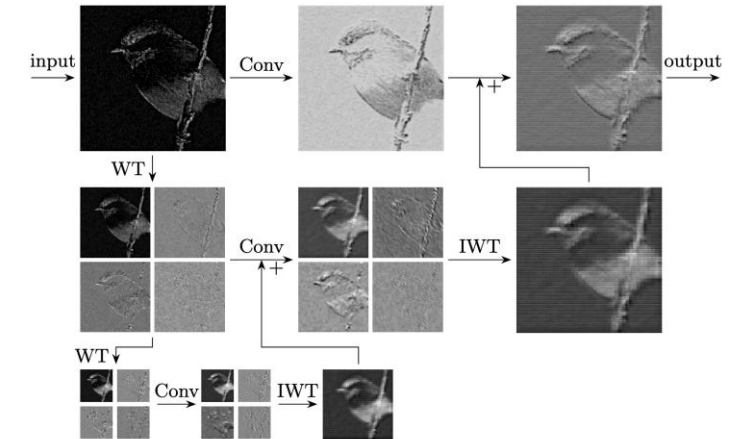


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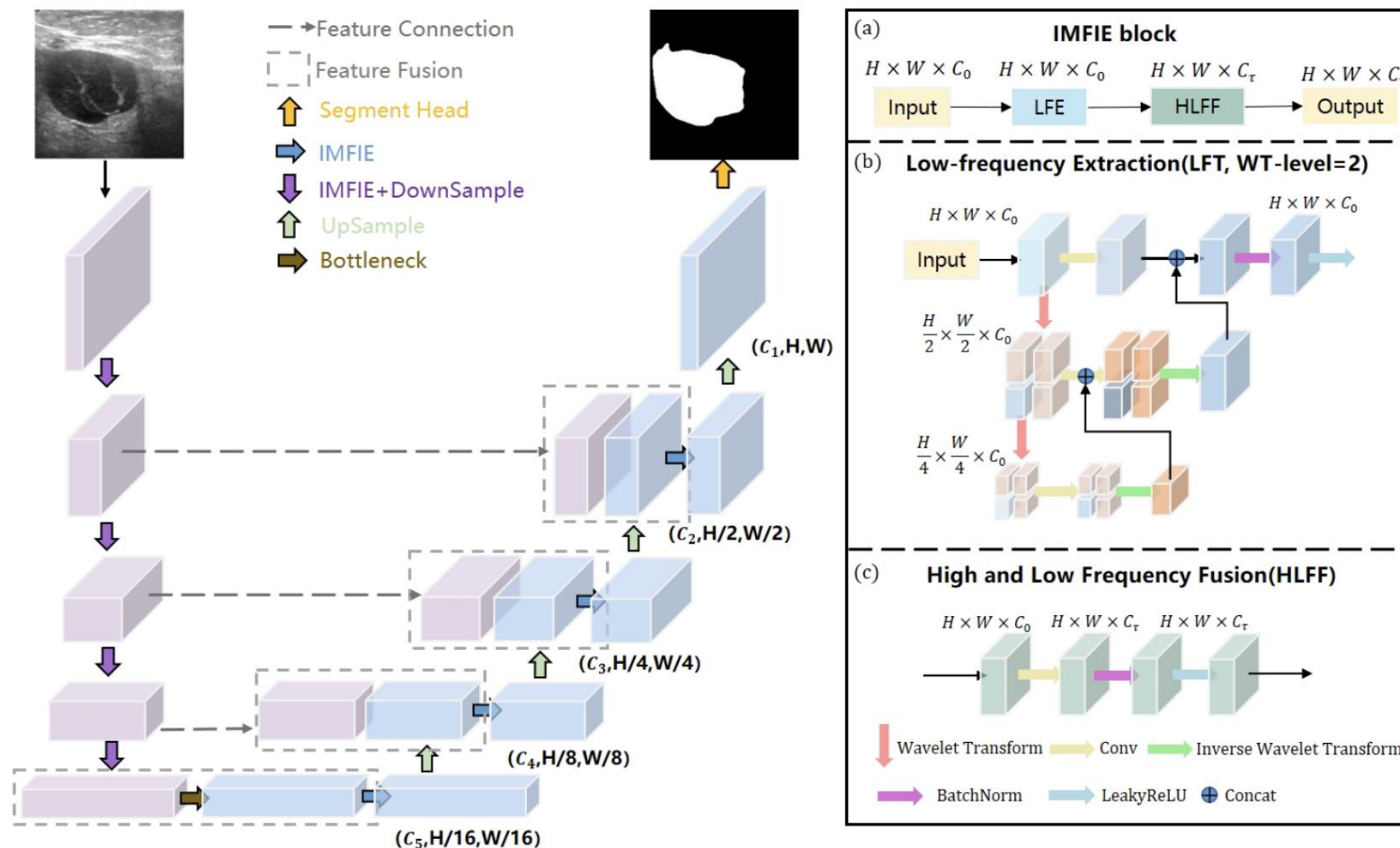


Fig. 2. Overview of UWT-Net. (a) IMFIE block, which contains Low-frequency Extraction (LFE) and High and Low Frequency Fusion (HLFF); (b) LFE mainly consists of Wavelet Convolutions, BatchNorm, and LeakyReLU; (c) HLFF is performed by standard convolution, BatchNorm, and LeakyReLU.

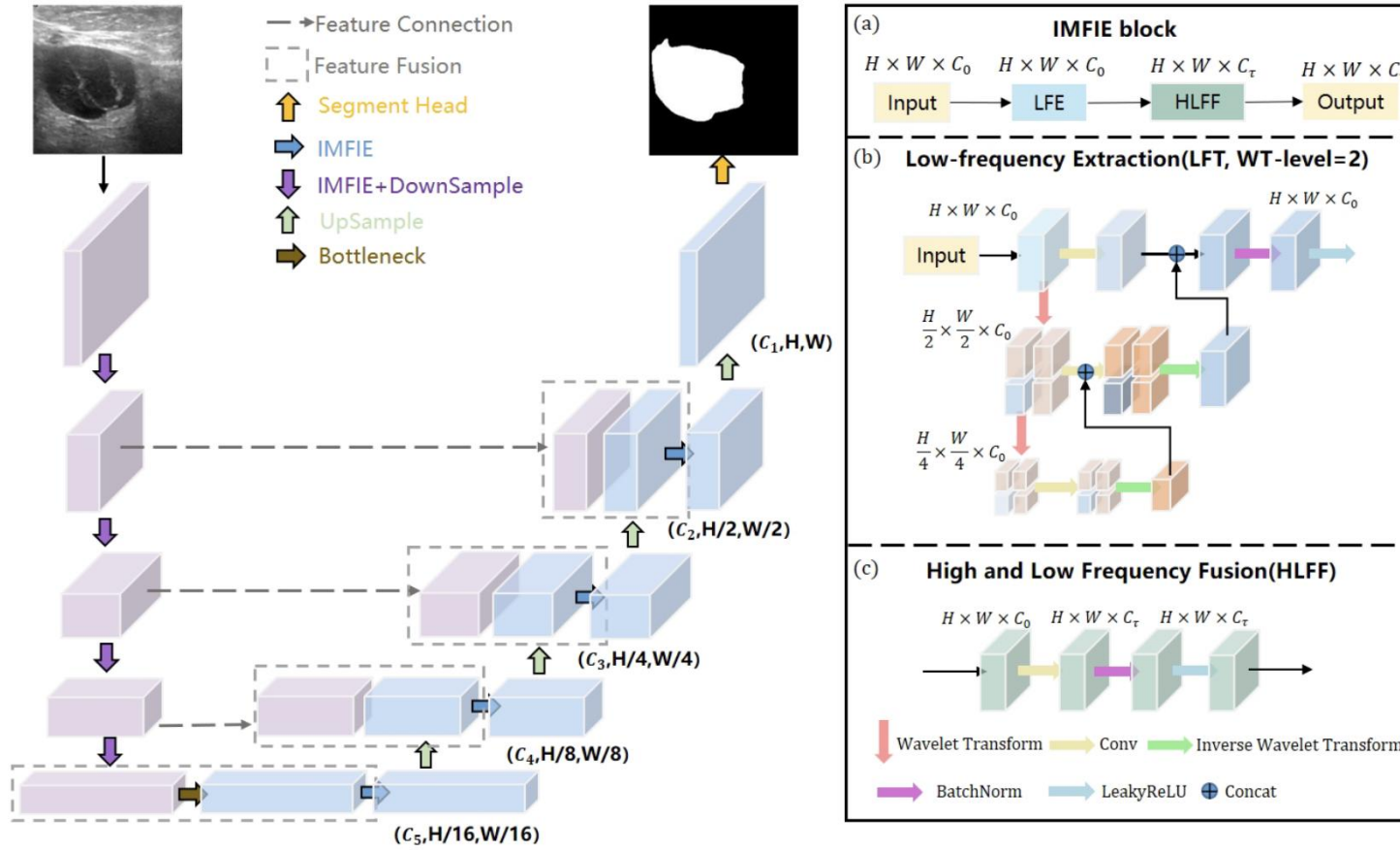


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$$I \in \mathbb{R}^{H_0 \times W_0 \times C_0}$$

$$O_\tau \in \mathbb{R}^{H' \times W' \times C_\tau}$$

$$O = LR(BN(WTConv(I)))$$

$$O_\tau = LR(BN(Conv(O)))$$

$$E_\tau = LR(MP(IMFIE(E_{\tau-1})))$$

$$D_\tau = Cat(E'_\tau, D'_\tau)$$

$$D_{\tau-1} = LR(US(IMFIE(D_\tau)))$$

$$D_0 \in \mathbb{R}^{H_0 \times W_0 \times C_Y}$$

$$L_{seg} = CE(T, UWT - Net(I))$$

- Datasets: BUSI, GlaS, CVC-ClinicDB
- Optimizer: Adam
- Learning rate: $1e-3$
- Scheduler: Cosine Annealing (minimum lr: $1e-4$)
- Loss: BCE+Dice
- Epoch: 400
- Metrics: IoU, F1 score
- Augmentation: random { rotation, flipping }
- Batch size: 8
- Train:Validation=8:2
- From scratch

Table 1. Performance comparison of various methods on BUSI, GlaS, CVC datasets, and average results.

Methods	BUSI		GlaS	
	IoU	F1	IoU	F1
U-Net [15](MICCAI'15)	63.90±0.65	77.50±0.55	87.49±0.89	93.04±0.61
Att-Unet [13](MIDL'18)	64.19±0.38	78.26±0.34	88.05±0.17	93.42±0.09
U-Net++ [21](MICCAI'18)	57.41±4.77	72.11±3.90	87.07±0.76	92.96±0.44
U-NeXt [17](MICCAI'22)	60.19±0.72	74.94±0.59	84.32±0.34	91.48±0.20
Rolling-UNet [11](AAAI'24)	63.74±0.32	77.64±0.19	87.74±0.18	93.46±0.10
U-Mamba [12](arXiv'24)	61.81±3.24	75.55±3.01	87.01±0.39	93.02±0.24
U-KAN [10](AAAI'25)	63.38±2.83	76.40±2.90	87.64±0.32	93.37±0.16
UWT-Net (Ours)	66.57±0.56	79.65±0.56	88.26±0.29	93.76±0.16

Methods	CVC		Average	
	IoU	F1	IoU	F1
U-Net [15](MICCAI'15)	83.71±0.48	91.05±0.31	78.36±0.67	87.20±0.49
Att-Unet [13](MIDL'18)	83.57±0.54	90.94±0.32	78.60±0.36	87.54±0.25
U-Net++ [21](MICCAI'18)	84.61±1.47	91.53±0.88	76.36±2.33	85.53±1.74
U-NeXt [17](MICCAI'22)	74.25±0.54	84.92±0.35	72.92±0.53	83.78±0.38
Rolling-UNet [11](AAAI'24)	81.31±0.78	89.48±0.49	77.60±0.43	86.86±0.26
U-Mamba [12](arXiv'24)	84.79±0.58	91.63±0.39	77.87±1.47	86.73±1.25
U-KAN [10](AAAI'25)	85.05±0.53	91.88±0.29	78.69±1.27	87.22±1.15
UWT-Net (Ours)	86.08±2.46	92.46±1.50	80.30±1.10	88.62±0.74

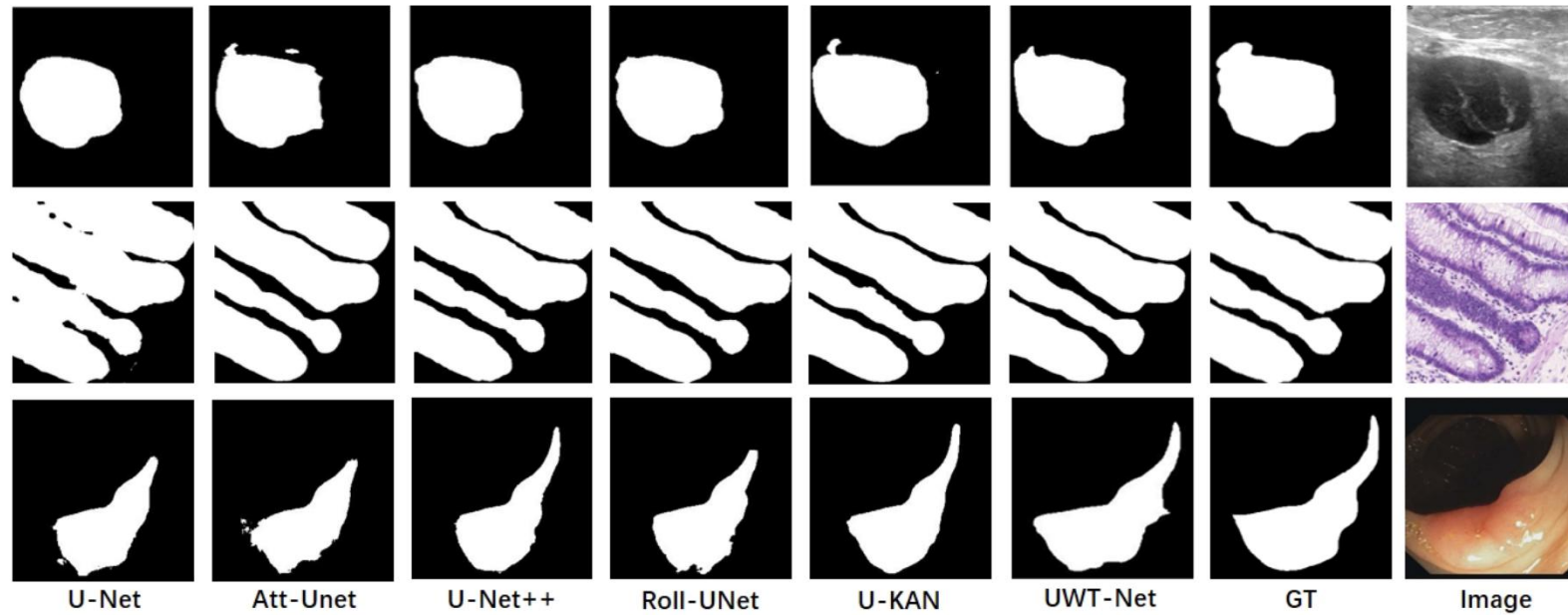


Fig. 3. The results of the proposed UWT-Net against all the compared methods over all the benchmarking datasets.

Table 2. Performance with and without IMFIE block across different datasets.

Feature extraction block	BUSI		GlaS		CVC	
	IoU	F1	IoU	F1	IoU	F1
standard convolution	62.28	75.86	83.58	91.37	80.11	88.72
IMFIE	67.14	80.17	88.54	93.92	87.63	93.38

Table 3. Performance with different WT-levels across average segmentation results.

WT-levels	Average	
	IoU	F1
Level=1	79.56±0.83	88.23±0.54
Level=2	80.30±1.10	88.62±0.74
Level=3	80.02±0.60	88.46±0.36

Table 4. Performance comparison with different model scales across average segmentation results.

Model Scale	Average		Efficiency	
	IoU	F1	Gflops	Params(M)
UWT-Net-S	79.38±1.08	88.02±0.75	8.37	7.71
UWT-Net	80.30±1.10	88.62±0.74	32.44	29.54
UWT-Net-L	80.61±0.36	89.03±0.28	127.70	155.56

- UWT-Net-S channels
 - [32, 64, 128, 256, 512]
- UWT-Net channels
 - [64, 128, 256, 512, 1024]
- UWT-Net-L channels
 - [128, 256, 512, 1024, 2048]

- In this study, we explore the potential of underutilized low-frequency features in medical imaging and propose an innovative block IMFIE and a novel model, UWT-Net.
- We perform empirical evaluations of our method under three medical image segmentation tasks.
- The results demonstrate that our model can effectively mine and utilize the overlooked low-frequency feature information and fuse it with the high-frequency information in medical images, improving segmentation effects.
- UWT-Net has the potential to be used in a wide range of medical image segmentation tasks.

- Contributions

- Low-frequency information contains spatial information like overall shape.
- Using WTConv CNNs can deal with global information.
- And it is also robust to noise.

- Limitations

- There are no comparisons with other WT based method like Spectral U-Net, Wet-unet.
- The authors experimented wide range of medical domain but there are no results which is related with CT or MRI.
- WT-IWT process is sequential it could be time consuming.

Please list the major weaknesses of the paper. Please provide details: for instance, if you state that a formulation, way of using data, demonstration of clinical feasibility, or application is not novel, then you must provide specific references to prior work.

- Limited Comparison to Related Wavelet-Based Methods: the paper does not provide a direct comparison to recent wavelet-based architectures, e.g. WeT-UNet (<https://pubmed.ncbi.nlm.nih.gov/38567422/>), or even different Wavelet-based. Kernels
- Clarity and Language Quality: The overall text clarity and English writing quality could be improved. The technical description and the current written style/language errors can potentially hinder its readability.
- Equation and Module Description: The mathematical explanation could be more rigorous and better structured. The flow of notation and formulation makes the text sometimes a bit inconsistent.
- Figure Quality and Captions: The captions and visualizations of Figures 1 and 2 can be improved. For instance, Figure 1 could explicitly label the LL, LH, HL, and HH wavelet subbands, and Figure 2 could depict the IMFIE block in a more modular and visually informative manner.
- External Validation: All evaluations rely on internal dataset splits. Including an external or cross-institution test set would better support claims of generalizability, e.g Kvasir for Polyp-baseb segmentation.